DualFace-Forgery: A Benchmark Dataset for Dual-Face Collaborative Forgery Detection

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Abstract. Recent advancements in deepfake generation technologies have significantly exacerbated security threats in face-centric social interaction scenarios. However, prevailing face forgery detection datasets and methodologies predominantly target single-face manipulation scenarios, failing to address the unique challenges posed by dual-face collaborative forgeries. Such forgeries exhibit amplified artifacts induced by interpersonal interactions and introduce intricate contextual inconsistencies that are absent in single-face cases, rendering existing detectors ineffective. To bridge this critical research gap, we present the Dual-Face Forgery Dataset (DFFD), the first comprehensive benchmark dataset exclusively tailored for dual-face collaborative forgery detection. DFFD comprises over 10,000 high-resolution images, where dual-face forgery samples are synthesized using state-of-the-art multimodal generative models based on real single-face images sourced from public datasets, with standardized preprocessing and fine-grained semantic annotations. Extensive experimental evaluations are conducted on four representative face forgery detectors (Xception, F3net, Recce, and Effort). The results demonstrate that detectors trained on DFFD effectively capture the generalizable underlying patterns of dual-face forgeries, yielding substantial improvements in cross-scene generalization performance and mitigating the inherent weak generalization limitation of existing detectors. Furthermore, we achieve accurate attribution of forged content to their respective generative models with a pre-trained classifier to enable forgery source traceability. This benchmark dataset enables pioneering research in social scene forgery forensics and provides essential foundational resources for the development of robust multi-face verification and authentication systems.

Keywords: Dual-Face Forgery Detection · Benchmark Dataset · Forgery Source Traceability.

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1 Introduction

The development of generative artificial intelligence (AIGC) technologies has enabled the production of hyper-realistic AI-synthesized group photographs. While these technologies offer unprecedented creative flexibility, their malicious exploitation has precipitated severe cybersecurity and societal risks. Malicious actors leverage forged group portraits to perpetrate financial fraud, identity impersonation, extortion, and disinformation campaigns, inflicting substantial financial losses and psychological harm on victims. Accordingly, the development of accurate and robust detection methodologies for forged group photographs has emerged as a critical and pressing research imperative.

Despite significant advancements in single-face forgery detection over the past decade [5,20,21,36], the detection of forged group photographs in unconstrained social interaction scenarios remains a largely unsolved challenge. Existing detectors exhibit degraded detection accuracy, limited robustness to post-processing operations, and poor cross-domain generalization capability, rendering them unsuitable for deployment in real-world operational environments. Compounding this challenge is the acute scarcity of publicly available benchmark datasets dedicated to group photograph forgery, which hinders the systematic training and rigorous evaluation of specialized detection models.

Prior efforts to enhance the generalization of face forgery detectors have predominantly centered on model regularization techniques and architectural innovations [2,8,19,23,26], with optimization strategies exclusively targeting the detector architecture rather than the training data distribution. Departing from this conventional paradigm, we propose a data-centric approach to address the generalization challenge by constructing a specialized benchmark dataset [7]. Specifically, we introduce the Dual-Face Forgery Dataset (DFFD), a large-scale benchmark characterized by multiple generative models, diverse forgery patterns, and comprehensive semantic annotations, tailored explicitly for dual-face collaborative forgery detection tasks. Training detectors on DFFD enables the model to learn generalizable underlying patterns of face forgery rather than overfitting to model-specific artifacts, thereby significantly enhancing cross-scene and cross-dataset generalization performance. Through extensive empirical evaluations, we demonstrate that detectors trained on DFFD exhibit superior robustness and generalization capability in cross-scene detection scenarios.

The main contributions of our paper are summarized as follows:

- We propose DFFD, the first comprehensive public benchmark dataset dedicated to dual-face collaborative forgery detection. This dataset fills a critical gap in the literature and provides a foundational resource for advancing research in group photograph forgery forensics.
- We empirically validate that training on DFFD yields substantial improvements in cross-scene detection performance across four detectors (Xception, F3net, Recce, and Effort) [3,27,32,33] , enhancing their practical utility in real-world applications.

– We augment existing detectors with a multi-classification head, which enables accurate forgery source attribution, supporting multi-task forensic analysis requirements.

2 Related Work

This section systematically reviews two core research areas closely related to this work: face forgery benchmark datasets and forgery source traceability technologies. We first outline the evolution of existing datasets from single-face to multi-face scenarios, then analyze the technical paradigms and limitations of current traceability methods, and finally clarify the research gaps addressed by this paper.

2.1 Dual-face Forgery Datasets

Benchmark datasets serve as the foundational infrastructure for advancing face forgery detection research. Over the past decade, the rise of deepfake technologies has driven the development of numerous high-quality single-face forgery datasets, which have significantly propelled the performance of single-face detection models. Representative works include FaceForensics++ (FF++) [22], which contains 1,000 real videos manipulated by four mainstream forgery methods (Deepfakes, Face2Face, FaceSwap, NeuralTextures), and the DeepFake Detection Challenge (DFDC) [6] dataset, a large-scale dataset comprising over 100,000 videos with diverse post-processing operations. These datasets have established standard evaluation protocols for single-face forgery detection and enabled the training of state-of-the-art detectors. However, as AI-generated group photographs have become increasingly prevalent in malicious applications, existing single-face-centric datasets have exposed critical limitations in addressing dual-face collaborative forgery scenarios. Current dual-face-related data sources can be broadly categorized into two types, both with significant shortcomings:

1) Dual-face subsets derived from single-face datasets: Datasets such as FF++ and DFDC contain only incidental dual-face samples, accounting for less than 5% of their total data volume. These samples are not specifically designed for collaborative forgery research, resulting in extremely limited forgery diversity—most are simple single-face transplants without considering interpersonal interaction contexts. Furthermore, they lack dedicated fine-grained annotations for dual-face scenarios, such as forged region localization and interaction consistency labels.

2) Real-world dual-face datasets: Datasets like PairHuman [17] focus on collecting natural dual-face interaction images with rich pose variations (e.g., mutual gaze, shoulder contact, and hand gestures) and high visual quality. However, they exclusively contain authentic images without corresponding AI-generated or tampered counterparts, making it impossible to form "real-fake paired" training data required for supervised detection models.

In summary, the field currently lacks a comprehensive benchmark dataset specifically tailored for dual-face collaborative forgery detection. Existing datasets

Table 1. Comparison of existing dual-face datasets

Comparison Dimension	Dual-face subsets (FF++, DFDC)	Real-world datasets (PairHuman)	Ours (DFFD)
Proportion of dual-face samples	Incidental	Sufficient	Sufficient
Sample design specificity	Not specific to dual-face forgery	Rich pose variations	Rich pose variations
Forgery scenario coverage	Simple single-face transplant	No forged samples	Multiple forgery scenarios
Annotation completeness	No dual-face specific annotations	No supervised annotations	Fine-grained annotations

suffer from insufficient sample size, limited forgery diversity, incomplete semantic annotations, and the absence of standardized real-fake pairing, which severely hinders the development and evaluation of specialized dual-face forgery detectors. A detailed comparison of existing dual-face datasets is shown in Table 1.

2.2 Traceability Detection Technology

Forgery source traceability aims to identify the specific generative model that produced a forged image, providing crucial evidence for digital forensics and judicial investigations. Existing traceability technologies are generally divided into two technical paradigms: active traceability and passive traceability.

Active Traceability Based on Digital Watermarking Active traceability techniques, predominantly represented by digital watermarking [10, 18, 24], embed imperceptible identification signals into images during the generation process to encode provenance information such as model ID, generation timestamp, and author credentials. Industry initiatives such as Google's Content Authenticity Initiative (CAI) and Adobe's Content Credentials have promoted the standardization of watermark-based traceability systems. Despite their theoretical simplicity, active methods have inherent practical limitations. First, watermarks are inherently fragile: they are easily corrupted or erased by common post-processing operations such as lossy JPEG compression, geometric cropping, filter enhancement, and noise addition, leading to irreversible traceability failure. Second, active methods require mandatory integration into generative model pipelines, rendering them completely ineffective for pre-existing unwatermarked content, which constitutes the vast majority of maliciously used forged images in real-world scenarios.

Passive Traceability Based on Generative Fingerprints To address the limitations of active methods, passive traceability techniques have emerged as the mainstream research direction. These methods operate without pre-embedded metadata by exploiting the intrinsic statistical artifacts—often referred to as "generative fingerprints" —inadvertently introduced by generative models during training [15,29,31,34]. These fingerprints manifest as consistent deviations in frequency domain distributions, gradient statistics, noise patterns, and texture correlations, which are unique to each model architecture and training configuration.

Early passive traceability methods relied on handcrafted statistical features, such as co-occurrence matrices and wavelet transform coefficients, to distinguish between real and AI-generated images. With the development of deep learning, CNN-based and Transformer-based methods have become dominant [1, 13], which automatically learn discriminative generative fingerprints from large-scale data. Recent advances have further explored multi-layer feature fusion and frequency-spatial joint learning to enhance the robustness of traceability against post-processing operations.

Nevertheless, existing passive traceability methods still face significant challenges in dual-face scenarios. First, most methods are designed and validated exclusively on single-face images, where the presence of two distinct faces introduces feature interference that confuses model judgments. Second, current research primarily focuses on traditional GAN-based and early diffusion models, with limited effectiveness against the latest multimodal large language model (MLLM)-based image generators [30], which produce more realistic images with subtler generative fingerprints. Finally, the lack of dedicated dual-face traceability datasets has prevented systematic evaluation and optimization of traceability models in group photograph scenarios.

3 DFFD Dataset

In this section, we introduce the proposed DFFD dataset, including raw data collection, data generation, data preprocessing, data annotation and dataset split.

3.1 Raw Data Collection

The raw data is collected by downloading real single-face images from the CelebFaces Attributes Dataset (CelebA) [14], which is a large-scale celebrity facial attribute dataset that contains over 200,000 celebrity images. It is widely used in computer vision research tasks, such as face recognition, facial attribute recognition, and pose estimation. The images feature rich variations in pose and background interference, providing high diversity.

A total of 4000 independent single-face images that meet the requirements of high definition and no facial occlusion are screened and finally determined from the CelebA dataset, which serve as the common source for all real and fake samples of the DFFD dataset. This design ensures that the two types of samples are in a homologous data distribution from the source, avoiding additional deviations introduced by differences in data sources.

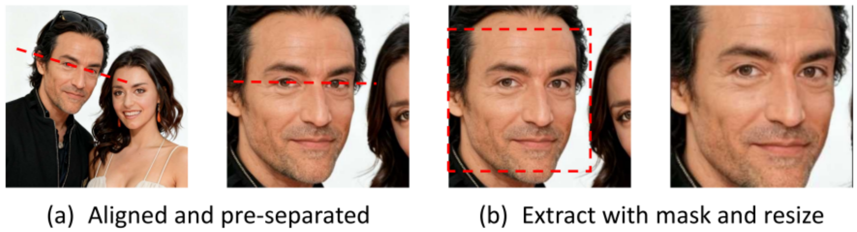

Fig. 1. The schematic diagram of the group photograph preprocessing process

3.2 Data Generation

The dual-face forgery group photographs are generated using multiple models: Doubao seedream 4.0 (hereinafter referred to as "Doubao"), GPT image 2.0 (hereinafter referred to as "GPT image"), Google nano banana 2.0 (hereinafter referred to as "Google nano"), and Alibaba tongyi wan 2.6 (hereinafter referred to as "Ali tongyi wan"), which are based on mainstream generative architectures such as Diffusion and Autoregressive (AR) [9, 25]. We call the API interfaces of these models to generate dual-face forgery group photographs in batches.

The generation pipeline takes the real single-face images selected from the CelebA dataset as base inputs. Combined with fine-grained prompt control that covers **three poses** (i.e., frontal view, upward view, mutual gaze) and **three expressions** (i.e., smiling, surprised, serious), our approach enables natural facial fusion of the two input identities and diversified output generation.

3.3 Data Preprocessing

The two-stage progressive processing approach achieves high-precision face separation and standardized output. In the first stage, the dlib 5-point landmark model aligns faces based on the horizontal baseline of the eye centers, preliminarily separating individual faces. In the second stage, the dlib 81-point landmark model precisely locates the full facial contour, including the forehead and jaw regions, and constructs a square mask based on the maximum dimension of facial width and height to extract faces [35]. This effectively avoids edge artifacts or feature loss commonly caused by traditional rectangular cropping. Finally, the cropped face regions are uniformly resized to 256×256 pixels, ensuring dimensional consistency and providing high-quality, standardized input for downstream tasks such as face recognition and feature extraction(see Fig. 1).

The highlighted regions in the Grad-CAM heatmap (see Fig. 2) denote the core feature areas that the model is interested in [16, 28]. For images in category (c) with residual faces, the highlights in the heatmap are predominantly located at the boundary between the two faces. This indicates that the model's attention is fragmented, preventing it from focusing on the holistic features of an individual face. As a result, the extracted feature vectors become distorted, which has an

Fig. 2. Comparison of Grad-CAM heatmaps for faces under different preprocessing methods

adverse impact on the accuracy of downstream tasks such as face recognition and expression analysis.

3.4 Data Annotation

The dual-face forgery group photographs derived from CelebA single-face samples are processed through unified preprocessing operations including facial alignment, face separation and resolution normalization, and the obtained single-face samples are labeled as "fake" . Meanwhile, the original real single-face samples directly from CelebA are processed through the exact same preprocessing pipeline, with the output samples labeled as "real" . The two types of samples not only share the homologous data distribution of CelebA, but also fundamentally eliminate domain artifacts introduced by inconsistent processing procedures via the fully aligned preprocessing workflow. This design effectively prevents subsequent detection models from achieving classification merely relying on irrelevant image statistical biases.

Furthermore, we generate a globally unique SHA-256 identity hash identifier for each independent source identity. Multi-dimensional metadata including the source identity hash, generative model origin and tampering method is bound to all samples, constructing a fully traceable dataset structure that provides complete underlying data support for downstream tasks such as forgery source tracing and fine-grained distribution analysis.

3.5 Dataset Split

After completing the full pipeline of data generation, preprocessing and annotation, the DFFD dataset is formally constructed, consisting of 4000 real samples and 6600 forgery sample across diverse tampering scenarios. The detailed statistical distribution of the samples is presented in Table 2.

The dataset is finally split into training, validation and test subsets at a ratio of 60%:15%:25%, following a strict source identity-level isolation rule. We first

Table 2. The detailed statistical distribution of the samples

Sample type	Method of obtaining		Sample size	Proportion(%)
Real sample	Single-face images from CelebA, preprocessing		4000	37.7
Fake sample	Dual-face forgery group photos, preprocessing		6600	62.3
	Model	**Prompt**		
	Doubao	Poses	1540	14.4
		Expressions	1500	14.1
	Google nano	Poses	600	5.7
		Expressions	580	5.5
	GPT image	Poses	600	5.7
		Expressions	600	5.7
	Ali tongyi wan	Poses	600	5.7
		Expressions	580	5.5

extract all unique identity hash identifiers for stratified sampling, assign all real samples and their derived forgery samples corresponding to the same identity hash entirely to a single subset, and subsequently conduct a full-set identity hash verification via set intersection operation to ensure no cross-subset identity overlap among the three splits. This design fundamentally avoids data leakage and guarantees that the reported performance can accurately reflect the cross-distribution generalization ability of the detection model. Additionally, a DFFD single dataset is constructed using forgery groups generated by the "Doubao" model, serving as a benchmark reference for single model sources.

4 Experiments

In this section, the performance of the detector trained by the DFFD dataset in terms of detection accuracy, generalization ability, and traceability ability is verified by testing the same domain data and cross-domain data.

4.1 Selection of Detectors and Evaluation Metrics

Four detectors, Xception, Recce, F3net, and Effort are selected for training and testing, and the area under the ROC curve (AUC) and accuracy (ACC) are used as performance evaluation metrics. The detectors are shown in Table 3.

4.2 Forgery Detection Performance Evaluation

First, the selected detectors are trained using the following three datasets: DFFD training set, DFFD _single training set (with fake data only generated by Doubao), and FF++ dataset (external dataset). After the training is completed, the detectors are tested as follows:

Table 3. Introduction to selected detectors

Detector	Core mechanism	Type	Backbone
Xception	Extreme Inception	Naive	Xception
Recce	Reconstruction-Classification Learning for Face Forgery Detection	Spatial	Xception
F3net	Fusion, Feedback and Focus for Salient Object Detection	Frequency	Xception
Effort	Orthogonal Subspace Decomposition for Generalizable AI-Generated Image Detection	OSD	ViT

- **Test on DFFD dataset:** The test data is from the DFFD test set, which verifies the performance of the detector within the DFFD data distribution.
- **Test on Extra-DFFD dataset:** The test data is generated by four models, namely Flux-2-pro (hereinafter referred to as "Flux pro"), Flux.Kontext-pro (hereinafter referred to as "Kontext pro"), Kling 2.1 (hereinafter referred to as "Kling"), and Jimeng 5.0 (hereinafter referred to as "Jimeng"), with a total of 800 test samples. Since these four models are completely different from the generative models used in DFFD, this test aims to verify the generalization ability of the detector under different data distributions.

The test results are shown in Tables 4 and 5, respectively. The results show that the detectors trained on the traditional public dataset FF++ have a significant performance bottleneck when transferred to the task of dual-face forgery detection. The average ACC and AUC for this type of sample are only about 50%, which is equivalent to random guessing and has no practical detection capability at all. Thus, they cannot meet the demand of dual-face forgery detection.

After being trained with the DFFD dataset constructed in this work, the performance of the detectors has been significantly improved. Compared with the detector trained on FF++, the average ACC and AUC are increased by 44% and 45.9%, respectively. Compared with the detector trained on fake data generated by the single Doubao generative model, the performance is also improved, it is because DFFD integrates fake samples generated by multiple models, enabling the detectors to learn general discriminative features of dual-face forgery during training on multi-distribution fake data, instead of overfitting to the specific artifacts of a single generative model.

The test results on Extra-DFFD data show that even for dual-face forgery samples generated by generative models that have never been exposed during the training stage, the detectors trained on the DFFD dataset still show excellent generalization performance, with an average ACC of 88.9% and an average AUC of 95.5%, which fully confirms that the proposed method has high practical application value in real-world dual-face forgery detection tasks.

Table 4. Test results on DFFD dataset (DFFD data from Doubao, Google nano, GPT image, and Ali tongyi wan; DFFD_single data from Doubao)

	Train set: DFFD		Train set: DFFD_single		Train set: FF++	
Detector	ACC	AUC	ACC	AUC	ACC	AUC
Xception	0.978	0.986	0.965	0.980	0.510	0.521
Recce	0.878	0.972	0.835	0.954	0.475	0.504
F3net	0.988	0.990	0.978	0.980	0.487	0.516
Effort	0.885	0.945	0.846	0.932	0.496	0.513
Average	0.932	0.973	0.906	0.962	0.492	0.514

Table 5. Test results on Extra-DFFD dataset (Extra-DFFD data from Flux pro, Kontext pro, Kling, Jimeng)

	Train set: DFFD		Train set: DFFD_single		Train set: FF++	
Detector	ACC	AUC	ACC	AUC	ACC	AUC
Xception	0.946	0.967	0.932	0.960	0.497	0.521
Recce	0.823	0.942	0.812	0.936	0.467	0.496
F3net	0.937	0.972	0.930	0.968	0.485	0.514
Effort	0.848	0.938	0.822	0.917	0.478	0.499
Average	0.889	0.955	0.874	0.945	0.482	0.508

4.3 Forgery Type Tracing Evaluation

Classification detector: it is composed of an Xception model as the backbone network and multiple classification detector heads [4,11]. Use the DFFD training set to train the Xception classification detector and then use the trained detector to test the data of the DFFD test set. The Macro F1 score, the Macro AUC, and the confusion matrix are selected as the evaluation metrics of traceability ability. The experimental results are shown in Fig. 3 and Table 6. Experimental results show that the classification traceability detector trained on the DFFD dataset achieves an average F1-score and AUC of 98%. As verified by the results of the confusion matrix, only a few misclassifications exist, and the general precision of traceability reaches a high level.

Table 6. Evaluation results of forgery type tracing performance

Model	Precision	Recall	F1-score	AUC
Doubao	0.99	0.99	0.99	0.992
Google nano	0.99	0.97	0.98	0.986
GPT image	0.98	0.99	0.99	0.990
Ali tongyi wan	0.96	0.98	0.97	0.984
Macro Average	0.98	0.98	0.98	0.988

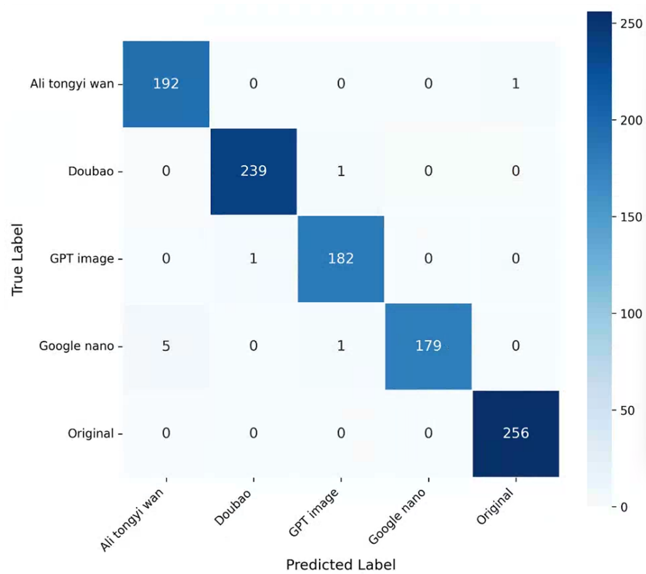

Fig. 3. The confusion matrix

5 Conclusion and Discussion

In this work, we have constructed a diverse dual face forgery dataset (DFFD) covering multiple forgery scenarios and multiple generative models. Experiments show that the face forgery detectors trained based on the DFFD dataset have achieved a significant improvement in forgery identification performance; even in the test scenario of cross-domain unseen datasets, the detectors still maintain excellent generalization ability. Further experimental verification shows that the classification traceability detector trained based on this dataset can achieve high-precision forgery source traceability.

However, the DFFD constructed in this work still has limitations: currently, the original images of both real and forged face samples in the dataset are all from the CelebA dataset, resulting in a relatively concentrated overall data distribution and insufficient coverage of the diversity of real-world scenarios.

Specific limitations can be verified through practical tests: when the trained detector is directly applied to the real face photograph scenarios publicly available on the Internet, since CelebA mostly contains cropped single frontal faces, the data distribution differs greatly from that of real group photographs. As a result, the accuracy of the detector in real dual-face group photograph scenarios drops by about 8-10 percentage points, and the generalization ability is significantly limited [12] .

To address this limitation, we propose a further optimization scheme: introduce the original real group photograph data from the PairHuman real group photograph dataset, first segment the face instances in the group photographs into independent single-face samples to construct a new "real" sample set; then use multiple mainstream generative models to generate dual-face forged group photographs based on the segmented real faces, and repeat the segmentation process to obtain a new "fake" sample set. After supplementing such multi-distribution samples, the adaptation ability of the dataset to practical application scenarios will be significantly improved.

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